YEAR 10

1) An object is launched from the bottom of pit 34.3 meters below the ground surface, directly upward at 39.2mls. The equation for the object's height *h* at time *t* seconds after launch is $h = -4.9t^2 + 39.2t - 34.3$. Where h is in meters, for how long is the object at or above the ground surface?

Factor completely. (Question 2-4)

2) $24xy^2 - 12xy + 36x^2y$.
3) $3ax^2 - 21axy + 30ay^2$.
4) $6m^2n^2 + 18m^2 n - mn^2$.
5) A kite is 32m above the ground. The angle the kite string makes with the ground 39^0. How long is the kite string to the nearest meter?

From a class of 30 students, 12 enjoy cricket (C), 14 enjoy Netball (N) and 6 enjoyed both cricket and netball. (Question 6-8)

6) Illustrate this information in a venn diagram.
7) State the number of students who enjoy (i) netball only (ii) neither cricket nor netball.
8) Find the probability that a person chosen at random will enjoy: (i) netball (ii) netball only (iii) both cricket and netball.

The area of rectangle lot is $3x^2 + 8x + 4$ (Question 9-10)

9) Determine the dimension of the lot
10) If the of value x is 4meters, state the dimensions and the area of the lot.

Complete the square for each of the following; then state the vertex for each. (Question 11-12)

11) $y = x^2 + 4x - 3$
12) $y = -3x^2 - 6x + 2$
13) Express $(3x^2 y^4)(4x y^2)$ in its simplest form.

14) Mark and Matthew's teacher has asked that they use their own rule to construct a sequence of number, starting with 5. The sequence that they have constructed are given below.

Matthew's sequence: 5; 9; 13; 17; 21: …
Mark's sequence: 5; 125; 3125; 78125; 1953 125: …

Write down the nth term (or the rule in terms of n) of:

- Matthew's Sequence
- Mark's sequence

The following sequence of numbers forms a quadratic sequence; -3; -2; -3; -6; -11; …... (Question 15-16)

15) Determine an expression for the n^{th} term of the quadratic sequence.

16) Calculate the first difference between the 35^{th} and 36^{th} terms of the quadratic Sequence.

Solve the following quadratic equations by factorizing. (Question 17-18)

17) $a^2 - 6a + 9 = 0$

18) $\frac{x2}{2} + \frac{7x}{4} = 0$

Given $\sum_{r=0}^{99}(3t - 1)$

19) Write down the first three terms of the series.

20) Calculate the sum of the series.